Ben Robison

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Education

University of Washington – BS in Electrical and Computer Engineering June 2028 (Expected)

- GPA: **3.8**; Dean's List (six quarters); Triangle Fraternity; Tau Beta Pi Engineering Honor Society
- Coursework: VLSI I (in progress), Digital Design, Computer Architecture (Winter), VLSI II (Spring)

Skills

RTL/HDL: SystemVerilog, Verilog, VHDL; AXI4, APB3, Avalon-MM; RTL design, CDC, UVM, FPGA synthesis, STA
Programming: C, Python, Java, SQL, Bash, Git; Linux, embedded Linux
Hardware/EDA: Quartus Prime, TimeQuest, Efinity, ModelSim, Cadence Virtuoso, HSPICE, KiCad; oscilloscopes

Experience

Undergraduate Researcher, ACME Lab @ UW – Seattle, WA Sep 2026 – Present

- Designing a streaming normalization stage for an hls4ml Gaussian-localization model: a listener computes per-crop min & max statistics while a ping-pong buffer holds each 48×48 crop, sustaining 1 pixel/cycle.
- Adapted a pipelined VHDL Gaussian detector from 5 to 3 Gaussians for a six-person team of grad students and industry engineers; built a self-checking testbench.

FPGA/Embedded Systems Engineer, Husky Robotics @ UW – Seattle, WA Feb 2026 – Present

- Designed a DMA traffic-generator module that emulates a camera video stream; used it to validate DMA transfers end-to-end from software.
- Adapted and integrated APB3 (PWM control registers) and AXI interfaces between custom RTL and an Efinix Ti375 RISC-V SoC; helped bring up DMA access over the AXI path.
- Automated the rover's soil-sampling sequence in C on a Raspberry Pi, coordinating auger, lift, and collection-box actuation with time-of-flight fill detection and Hall-effect homing and dig-stop limits.
- Refactored the team's shared C codebase with conditional compilation and Makefile targets for reuse.

Summer Associate, rehired as Program Analyst, CBRE (Google Account) – San Jose, CA Jun 2026 – Sep 2026

- Evaluated two architectures for a SQL KPI dashboard; implemented one and won approval from CBRE and a senior Google stakeholder.

Projects

FPGA GEMM Accelerator ae0lis.github.io/projects/gemm_accelerator

- Designed an 8×8 signed-INT8 result-stationary GEMM accelerator in SystemVerilog using 64 parallel MAC units with INT32 accumulation; synthesized to 64 Cyclone V DSP blocks.
- Integrated the accelerator with the ARM HPS over Avalon-MM and wrote an embedded Linux C driver to load operands, control execution, and compare all 64 outputs against a software reference on a DE1-SoC.
- Computed one 8×8-by-8×8 matrix product in nine core cycles (180 ns at 50 MHz, excluding host transfers) with a 77.48 MHz post-fit worst-corner Fmax.
- Verified the RTL with self-checking testbenches and 1,000 randomized control-interface operations.

FPGA Rhythm Game ae0lis.github.io/projects/dwyt

- Built a four-lane rhythm game on a Cyclone V with per-row FSMs, LFSR note generation, and two-flop input synchronizers; reduced logic ∼35% to 330 ALUTs and 194 registers.

Leadership & Activities

Vice President & Assistant Instructor, UW Historical Fencing – Seattle, WA Sep 2025 – Present

- Helped revive the club: recruited officers, set up bank account, ran events, managed sponsors; grew first-meeting turnout from ∼12 to 30+.